INDIAN INSTITUTE OF TECHNOLOGY, KANPUR

DATA MINING PROJECT — MTH443

Patient Risk Segmentation

Analysis of the Diabetes Prediction Dataset

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This experience has been an extraordinary learning opportunity. It allowed us to apply theoretical knowledge from the course *MTH443 Data Mining* to a real-world clinical dataset, enhancing our understanding of exploratory data analysis, unsupervised learning, association rule mining, and supervised classification.

We also thank our fellow students for their constructive feedback and collaboration during the project .

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Abstract

This report presents an end-to-end data mining framework for clinical risk assessment using the **Diabetes Prediction Dataset** (Kaggle, 100,000 records). A rigorous preprocessing pipeline was applied: records with age below 18 were excluded to focus on adult-onset patterns, the `smoking_history` column was dropped due to high informational entropy (≈35% "No Info" entries), and duplicate records were removed. The final analytical cohort contained **82,767** unique adult observations, with a diabetic prevalence of approximately **10.17%**.

Exploratory Data Analysis (EDA) quantified key physiological divergence between diabetic and non-diabetic groups. Diabetic individuals exhibited notably higher mean blood glucose (+46%), HbA1c (+29%), BMI (+13%), and age (+29%) compared with non-diabetics.

To identify hidden patient subgroups without using the diagnostic label, K-Means clustering was applied using composite features inspired by the RFM framework. Here, Recency represents age, Frequency represents comorbidity count (hypertension and heart disease), and Monetary represents overall metabolic burden (BMI, HbA1c, and blood glucose).

The optimal choice of $k = 3$ successfully segmented the population into clinically meaningful low-risk, medium-risk, and high-risk groups. These clusters were primarily driven by differences in HbA1c and blood glucose levels.

The clustering results were further validated using hierarchical clustering and PCA-based visualisation, both of which confirmed clear and well-separated group structures.

Disease Basket Analysis using the Apriori algorithm identified strong co-occurrence rules among binary health indicators (High Glucose, High HbA1c, Obesity, Old Age, Hypertension, Heart Disease), with lift values reaching 1.687.

Seven supervised classification models—LDA, Logistic Regression, SVM, Decision Tree, CART, Random Forest, and MLP—were trained using SMOTE oversampling. **Random Forest** achieved the best overall AUC performance, with HbA1c and blood glucose identified as the two dominant predictive features.

Contents

1 Introduction

Diabetes mellitus is one of the most prevalent non-communicable diseases worldwide, placing an enormous burden on both individuals and healthcare systems. According to the World Health Organization, over 422 million adults currently live with diabetes, and the number is projected to rise sharply in coming decades. Early identification of at-risk individuals is therefore a critical public health priority.

In recent years, the integration of data mining techniques into healthcare analytics has significantly improved the ability to extract meaningful, actionable insights from large-scale clinical datasets. This project focuses on **Patient Risk Segmentation** using the Diabetes Prediction Dataset—a large clinical repository containing 100,000 records with demographic attributes, comorbid conditions, and key metabolic biomarkers such as BMI, HbA1c level, and blood glucose level.

The primary objectives of this study are:

- To identify hidden patterns and risk factors in patient data using Exploratory Data Analysis (EDA).
- To segment individuals into clinically meaningful risk groups using unsupervised learning (K-Means and Hierarchical Clustering).
- To discover co-occurring health conditions strongly associated with diabetes through Disease Basket Analysis (Apriori algorithm).
- To build and compare predictive classification models for early diabetes detection, evaluated under a cost-sensitive framework.

After preprocessing, EDA revealed clear physiological differences between diabetic and non-diabetic individuals. In particular, HbA1c and blood glucose levels showed the strongest separation between classes, consistent with their established clinical diagnostic role. K-Means clustering recovered three meaningful patient archetypes aligned with low-, moderate-, and high-risk metabolic profiles. Finally, among seven supervised models, Random Forest emerged with the highest AUC.

2 Dataset Overview

The dataset used in this study is the *Diabetes Prediction Dataset*, obtained from the Kaggle platform. It contains **100,000 patient records** with a combination of demographic, physiological, and medical attributes.

2.1 Dataset Composition

The dataset consists of **9 features** classified into four functional domains:

- **Demographic Features:** gender, age
- **Comorbidity Indicators:** hypertension, heart_disease
- **Physiological Measurements:** bmi, HbA1c_level, blood_glucose_level
- **Behavioral Feature:** smoking_history (excluded after preprocessing)
- **Target Variable:** diabetes (1 = Diabetic, 0 = Non-diabetic)

2.2 Feature Description

Table 1: Overview of Dataset Features

Feature	Type	Description
gender	Categorical	Biological sex of the patient
age	Numeric	Age in years (continuous)
hypertension	Binary	Presence of hypertension (1 = Yes, 0 = No)
heart_disease	Binary	History of heart disease (1 = Yes, 0 = No)
smoking_history	Categorical	Smoking status (excluded in analysis)
bmi	Numeric	Body Mass Index (kg/m^2)
HbA1c_level	Numeric	Glycated haemoglobin (%); 3-month glucose average
blood_glucose_level	Numeric	Fasting blood glucose concentration (mg/dL)
diabetes	Binary	Target variable (1 = Diabetic, 0 = Non-diabetic)

2.3 Class Distribution

The dataset exhibits a significant class imbalance:

- **Non-Diabetic (class 0):** $\approx$ 91.5% of records
- **Diabetic (class 1):** $\approx$ 8.5% of records

This severe imbalance necessitates the use of appropriate evaluation metrics (Precision, Recall, F1-score, AUC-ROC) rather than simple accuracy, and the application of resampling techniques such as SMOTE during model training.

3 Computational Pipeline: Data Pre-processing

3.1 Preprocessing Steps

The following sequential operations were implemented:

1. **Age Filtering:** Records with age < 18 were removed (focus on adult-onset Type-2 diabetes). This pruned 17,201 records.
2. **Feature Removal:** The `smoking_history` column was dropped due to ≈35.82% "No Info" values.
3. **Deduplication:** Duplicate rows were removed, yielding the final cohort of **82,767** unique records.
4. **Feature Engineering:**
   - `age_group`: Age discretised into bins [0,18,30,45,60,80] with labels Child/Young/Adult/Middle/Senior.
   - `age_group_encoded`: Ordinal integer encoding (Child=0, Young=1, Adult=2, Middle=3, Senior=4).
   - `gender_encoded`: Male=0, Female=1.

3.2 Post-Processing Cohort Summary

Analytical Cohort Summary

Metric	Description	Value
Final Sample Size	Unique adult observations after all steps	**82,767**
Diabetic Prevalence	Positive class (`diabetes` = 1)	**≈10.17%**
Features for Modelling	After encoding and dropping extras	**8**
Missing Values	After preprocessing	**0**

4 Exploratory Data Analysis

EDA was performed to understand the structure, distribution, and inter-variable relationships before applying machine learning techniques.

4.1 Class Distribution

The target variable `diabetes` shows severe class imbalance (≈90:10 ratio). This was addressed using SMOTE during model training. Evaluation relied on Recall, F1-score, and AUC-ROC rather than raw accuracy.

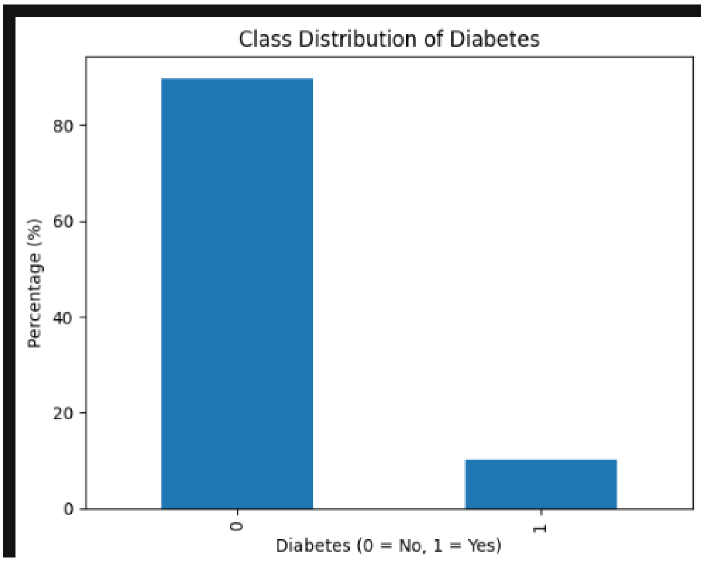


Figure 1: Class distribution of the `diabetes` target variable. Approximately **90%** non-diabetic (class 0) and **10%** diabetic (class 1).

4.2 Multi-Panel Histograms by Diabetes Status

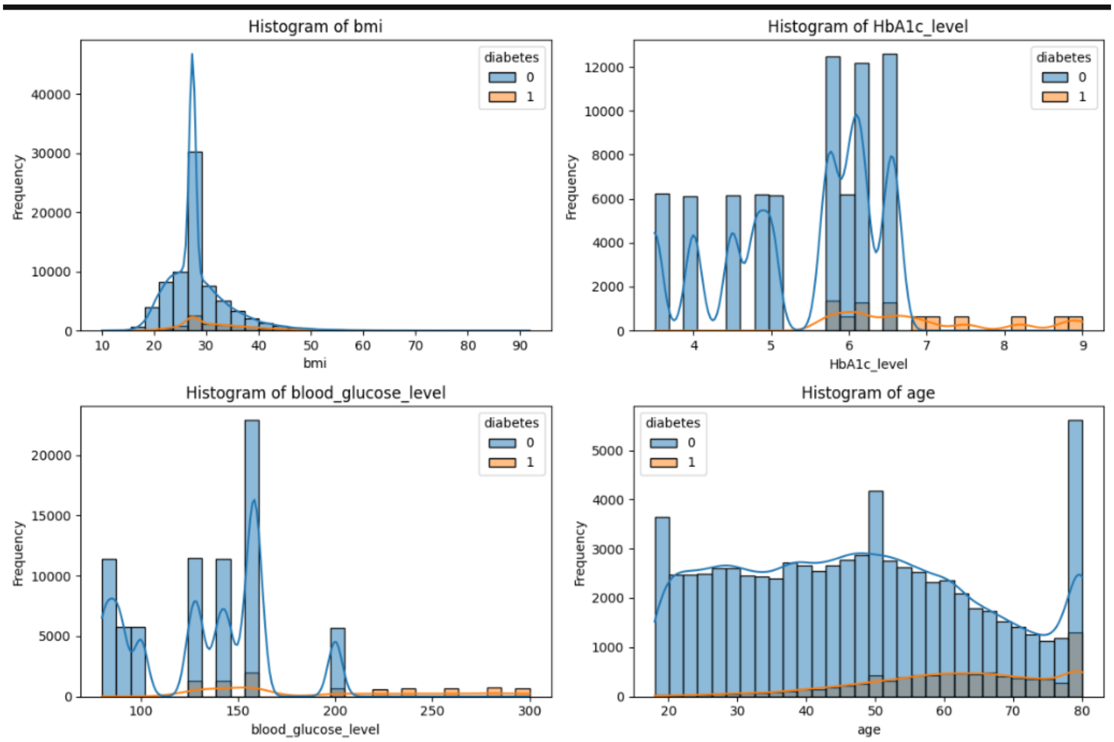


Figure 2: Multi-panel histograms with KDE for BMI, HbA1c, Blood Glucose Level, and Age, stratified by diabetes status (blue = Non-diabetic, orange = Diabetic). Diabetic patients cluster at higher glucose and HbA1c values.

Key observations:

- **BMI:** Right-skewed with peak near 27–28. Diabetics show a slightly elevated distribution.

- **HbA1c Level:** Multimodal; diabetic distribution shifts rightward, consistent with WHO diagnostic threshold $\geq 6.5\%$.
- **Blood Glucose:** Also multimodal. Diabetics cluster strongly at higher values (> 180 mg/dL).
- Blood glucose and HbA1c are the most discriminating features.

4.3 Boxplot Analysis

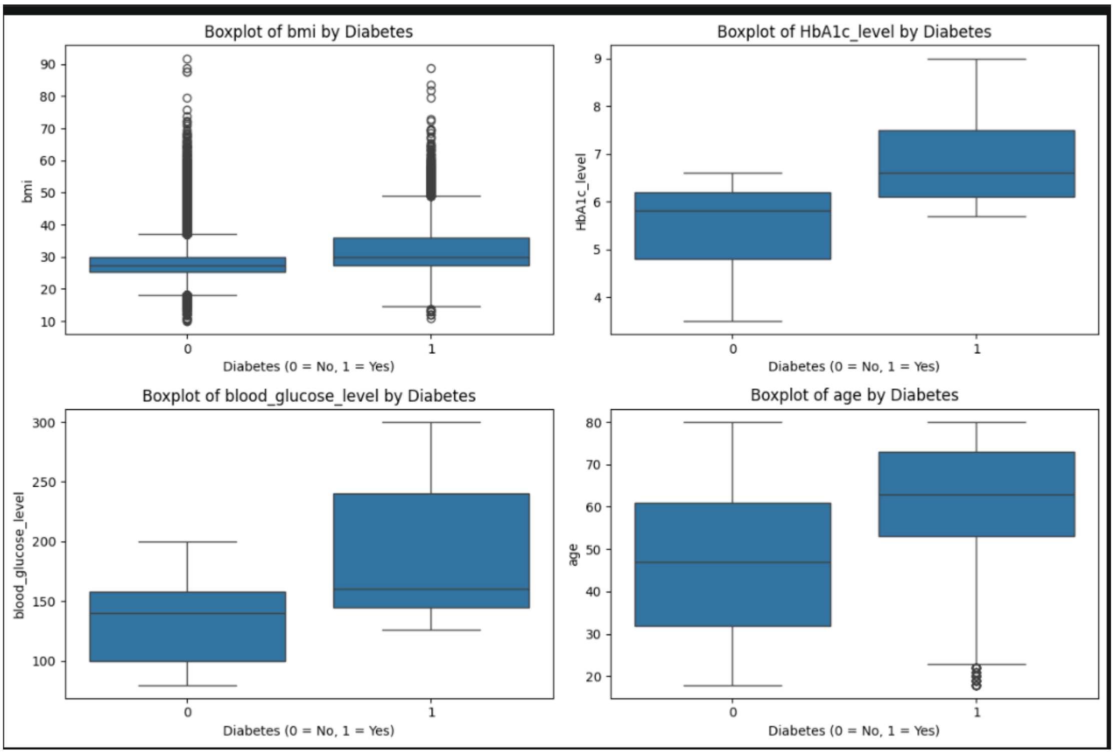


Figure 3: Combined 2×2 boxplots of BMI, HbA1c, Blood Glucose Level, and Age by diabetes status (0 = Non-diabetic, 1 = Diabetic).

The boxplots confirm:

- **Age:** Diabetics have median age ≈ 63 yrs vs. ≈ 47 yrs for non-diabetics.
- **BMI:** Diabetics show slightly elevated median (≈ 32) vs. non-diabetics (≈ 27).
- **HbA1c:** IQR for diabetics (6.5–7.5%) lies almost entirely above the non-diabetic median ($\approx 5.8\%$)—strongest visual discriminator.
- **Blood Glucose:** Diabetics show much wider spread (IQR $\approx$ 160–240 mg/dL) vs. non-diabetics (IQR $\approx$ 100–160 mg/dL).

4.4 Violin Plots

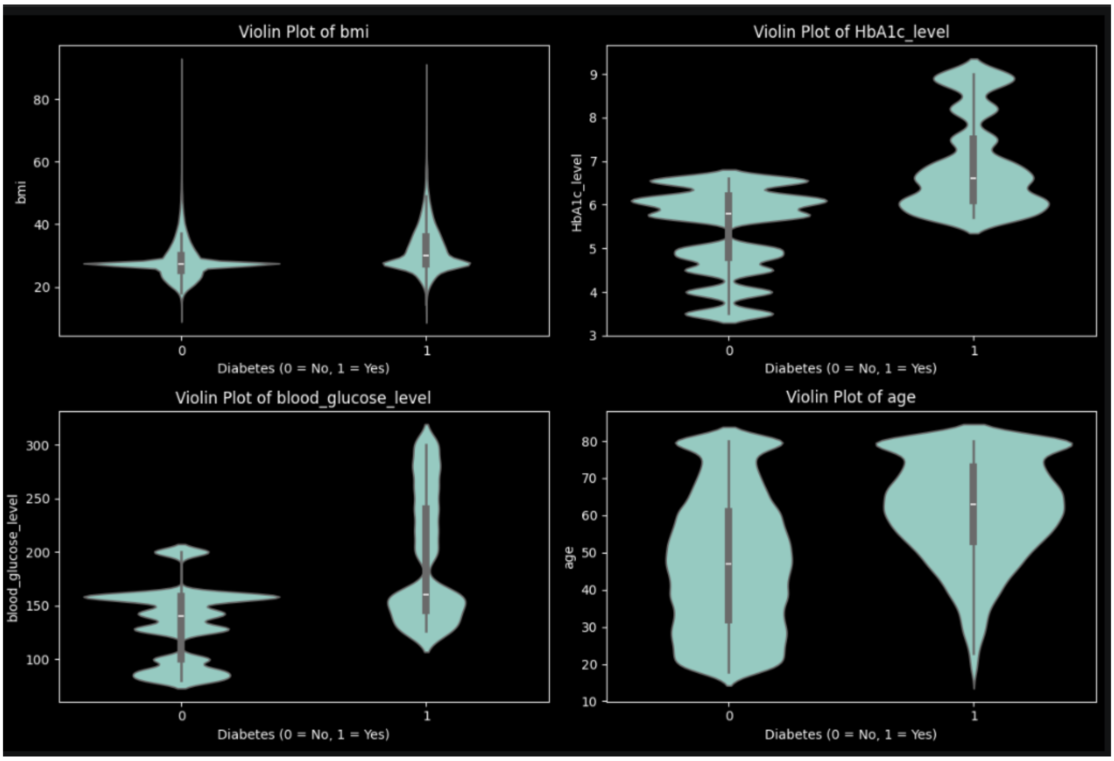


Figure 4: Violin plots of BMI, HbA1c, Blood Glucose Level, and Age stratified by diabetes status. Blood glucose and HbA1c show bimodal patterns for the diabetic group, confirming clinical cut-off thresholds.

Interpretation: The violin plots highlight clear differences between diabetic and non-diabetic individuals across all variables. BMI distributions are relatively similar, although diabetic individuals tend to have slightly higher values. In contrast, HbA1c levels are significantly higher in the diabetic group, with a wider spread indicating greater variability. Blood glucose levels show the most pronounced difference, with a clear separation between the two groups and much higher values among diabetic patients. Additionally, diabetic individuals are generally older, as seen in the age distribution. Overall, HbA1c and blood glucose emerge as the most important distinguishing features for diabetes.

4.5 Profile Plot

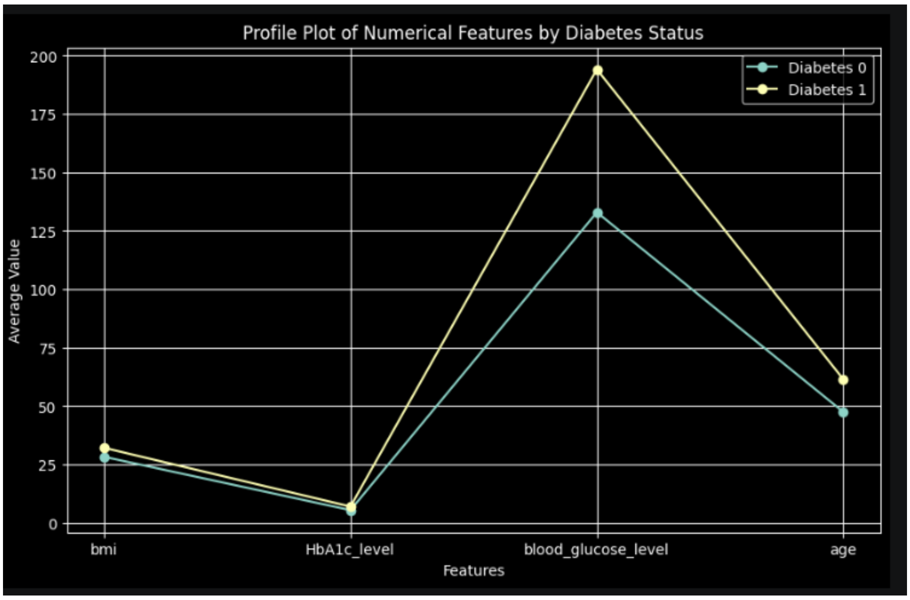
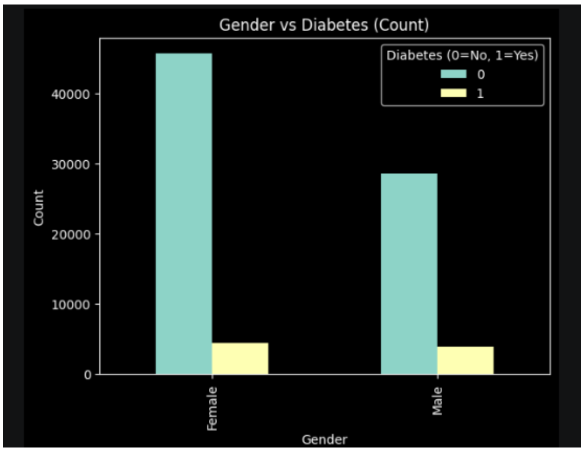


Figure 5: Profile plot showing the mean value of each numerical feature for diabetic vs. non-diabetic patients. Blood glucose level has the largest absolute gap between classes.

The profile plot confirms that **blood glucose level** has the largest absolute gap (≈ +46% for diabetics), followed by HbA1c (+29%), age (+29%), and BMI (+13%).

Interpretation: The plot clearly shows that diabetic patients have higher values across all features compared to non-diabetic individuals. The biggest difference is observed in blood glucose levels, indicating that it is the strongest indicator of diabetes. HbA1c also shows a significant increase, reinforcing its importance in diagnosis. Age differences suggest that diabetes is more common in older individuals, while BMI shows a smaller increase, indicating a weaker but still relevant effect. Overall, blood glucose and HbA1c are the most important factors distinguishing diabetic and non-diabetic patients.

4.6 Gender vs. Diabetes



(a) Count of diabetic vs. non-diabetic patients by gender.

Interpretation: The plot shows the number of diabetic and non-diabetic individuals across genders. It can be observed that the majority of patients in both groups are non-diabetic, which reflects the class imbalance in the dataset. While the total number of female patients is higher than male patients, the proportion of diabetic cases appears relatively similar for both genders. This suggests that gender alone is not a strong distinguishing factor for diabetes in this dataset, and other features such as HbA1c and blood glucose play a more significant role.

4.7 Age Group Analysis

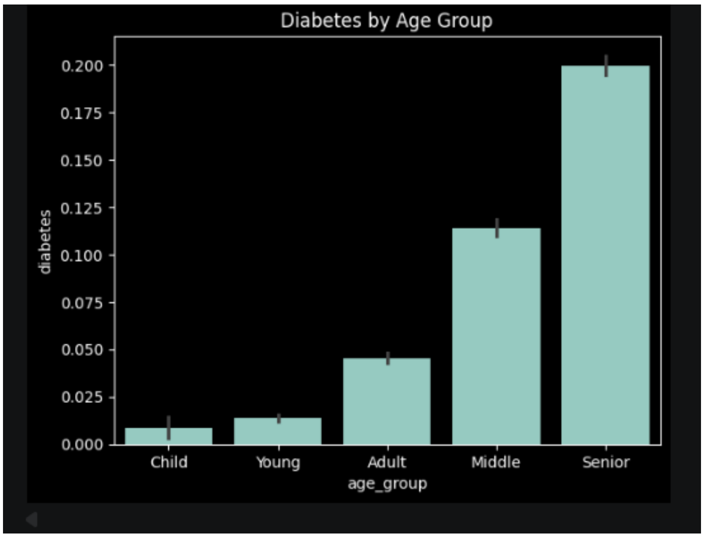
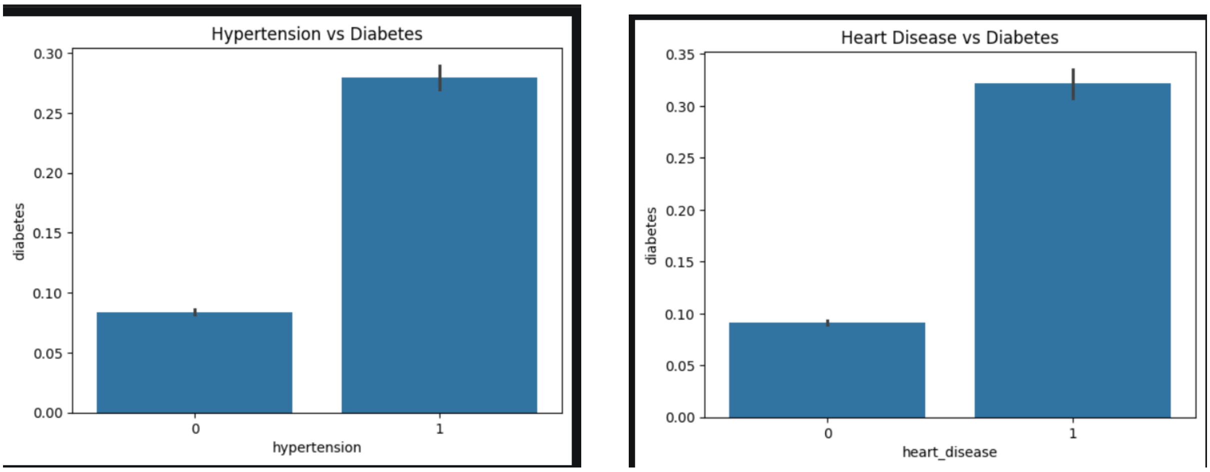


Figure 7: Diabetes prevalence across age groups. Prevalence rises monotonically from $< 2\%$ in Young individuals to $\approx 20\%$ in Seniors.

Interpretation: The figure shows a clear increasing trend in diabetes prevalence with age. Very few cases are observed in children and young individuals, while the proportion gradually increases in adults and becomes significantly higher in middle-aged and senior groups. The highest prevalence is seen among seniors, indicating that older individuals are more prone to diabetes. This suggests that age is a strong risk factor and plays an important role in diabetes development.

4.8 Binary Feature Analysis: Hypertension and Heart Disease



(a) Diabetes prevalence by hypertension status.

(b) Diabetes prevalence by heart disease status.

Figure 8: Comorbidity features show elevated diabetes risk: hypertension ($\approx 3.3\times$ higher relative risk) and heart disease ($\approx 3.8\times$ higher relative risk).

4.9 Chernoff Faces Visualisation

Chernoff Faces encode four normalised features into facial attributes:

- *Face width* $\leftarrow$ normalised **age**
- *Eye size* $\leftarrow$ normalised **BMI**
- *Mouth curve* $\leftarrow$ normalised **HbA1c level**
- *Nose length* $\leftarrow$ normalised **blood glucose level**

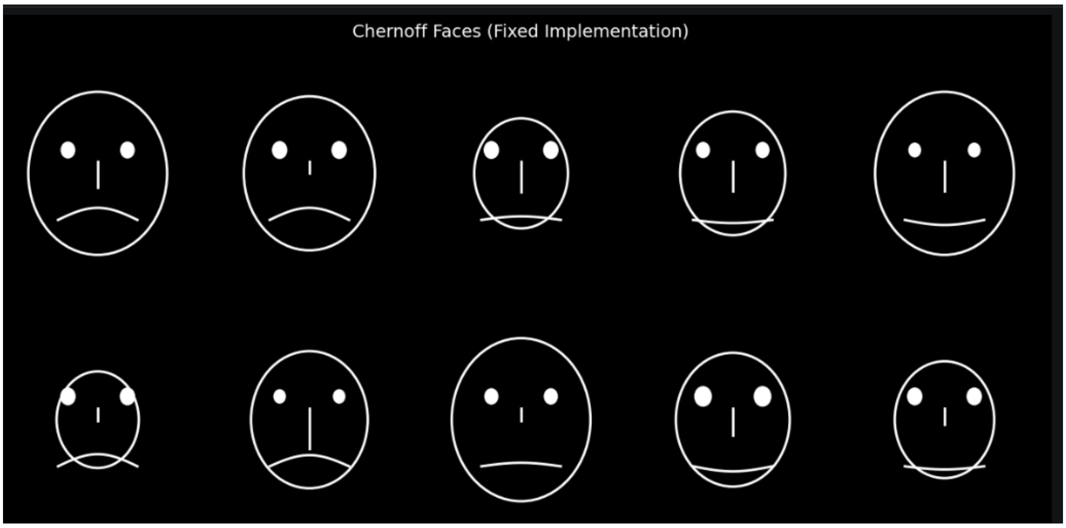


Figure 9: Chernoff Faces for 10 sample patients (dark background). Patients with high glucose and HbA1c tend to have longer noses and upward mouth curves.

Interpretation: The Chernoff faces provide a visual way to compare patients using facial features. Each face represents one patient, where changes in facial characteristics reflect differences in health indicators. Patients with longer noses correspond to higher blood glucose levels, while upward mouth curves indicate higher HbA1c levels. Wider faces represent older individuals, and larger eyes correspond to higher BMI values. By observing these faces, it becomes easier to identify patterns, as patients with more severe diabetic indicators tend to show noticeable facial differences compared to others. Overall, this visualization helps in quickly understanding how multiple features vary together across patients.

4.10 Correlation Matrix

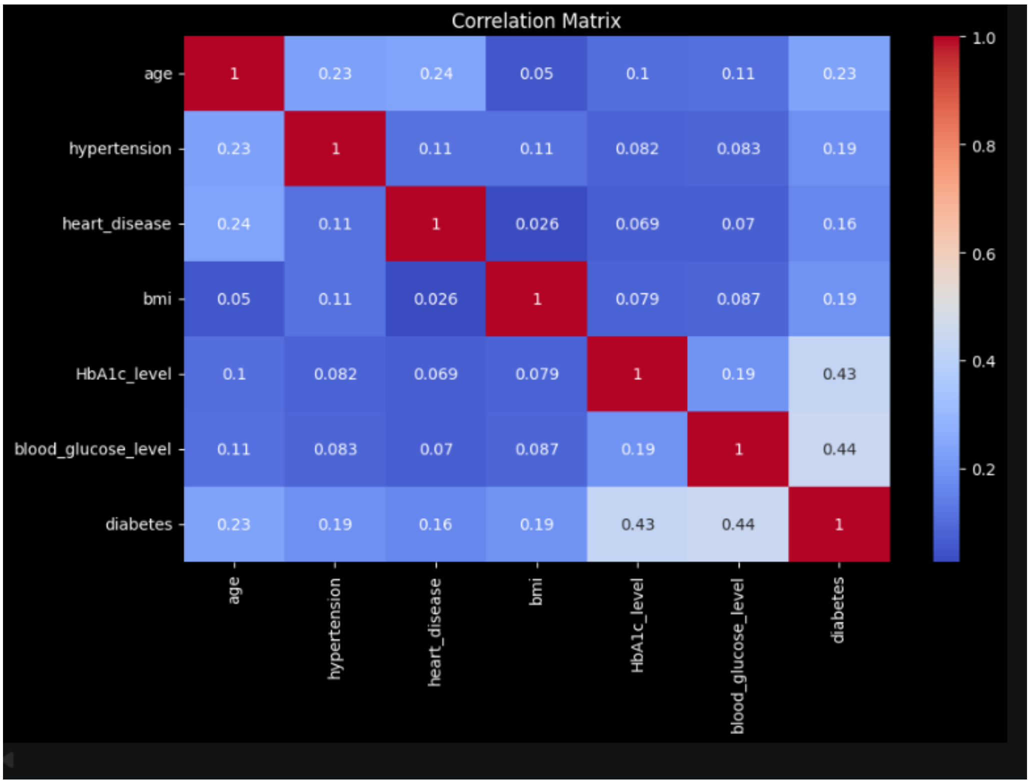


Figure 10: Pearson correlation heatmap (annotated). Red = positive correlation; blue = negative correlation.

From Figure 10, correlations with `diabetes` are:

- **Blood glucose level:** Highest correlation ($r = 0.44$).
- **HbA1c level:** Second highest ($r = 0.43$).
- **Age** ($r = 0.23$), **BMI** ($r = 0.19$), **Hypertension** ($r = 0.19$), **Heart disease** ($r = 0.16$) show moderate correlations.
- No pair exceeds $r > 0.8$, confirming absence of severe multicollinearity.

EDA Summary: Blood glucose and HbA1c are the most discriminating features. The severe class imbalance ($\approx 90 : 10$) was addressed via SMOTE in the modelling phase.

5 Analysis

5.1 Disease Basket Analysis (Apriori Algorithm)

Association rule mining was performed on a binary transaction representation. Binary features were engineered from continuous variables:

Binary Feature	Definition
`High_Glucose`	`blood_glucose_level` > 180
`High_HbA1c`	`HbA1c_level` > 6.5
`Obese`	`bmi` > 30
`Old_Age`	`age` > 50
`Hypertension`	`hypertension` $= 1$
`Heart_Disease`	`heart_disease` $= 1$

The **Apriori algorithm** (`mlxtend`) was applied with:

$$\text{min\_support} = 0.05, \qquad \text{min\_confidence} = 0.50$$

Top Rules by Confidence

Antecedents	Consequents	Support	Confidence	Lift
{High_HbA1c, Obese}	{Old_Age}	0.052	0.791	1.687
{Hypertension}	{Old_Age}	0.102	0.787	1.679
{High_HbA1c}	{Old_Age}	0.127	0.781	1.665
{High_Glucose, Hypertension}	{Old_Age}	0.057	0.777	1.657
{Obese, Hypertension}	{Old_Age}	0.069	0.774	1.651

Interpretation: The highest-confidence rules consistently point to `Old_Age` as the consequent. Individuals with elevated HbA1c, obesity, or hypertension are $\approx 79\%$ likely to be in the older-age group (> 50 years), with lift values ≈ 1.68.

Top Rules by Support

Antecedents	Consequents	Support	Confidence	Lift
{High_Glucose}	{Old_Age}	0.147	0.714	1.522
{Old_Age}	{High_Glucose}	0.147	0.313	1.522
{High_HbA1c}	{Old_Age}	0.127	0.781	1.665
{Old_Age}	{High_HbA1c}	0.127	0.271	1.665
{High_HbA1c, High_Glucose}	{Old_Age}	0.104	0.771	1.643

Top Rules by Lift

Antecedents	Consequents	Support	Confidence	Lift
{High_HbA1c, Obese}	{Old_Age}	0.052	0.791	1.687
{Hypertension}	{Old_Age}	0.102	0.787	1.679
{High_HbA1c}	{Old_Age}	0.127	0.781	1.665
{High_HbA1c, High_Glucose, Obese}	{Old_Age}	0.051	0.780	1.663
{High_Glucose, High_HbA1c}	{Old_Age}	0.104	0.771	1.643

Summary of Disease Basket Findings

- High blood glucose and high HbA1c are the strongest individual risk signals, frequently co-occurring with older age and obesity.
- Hypertension shows strong association with older age (confidence $\approx 79\%$, lift ≈ 1.68).
- Multi-condition baskets (e.g., High_HbA1c + Obese + High_Glucose) produce the most discriminative rules, confirming that co-occurring conditions are compounding risk factors.

5.2 Patient Stratification (Clustering)

5.2.1 RFM Feature Engineering

Three composite features inspired by the **RFM (Recency, Frequency, Monetary)** framework were constructed:

I. Recency ($\mathcal{R}$): Represented by `age`. Proxy for cumulative physiological exposure.

$$\mathcal{R}_i = \text{Age}_i$$

II. Frequency ($\mathcal{F}$): Comorbidity Accumulation Index.

$$\mathcal{F}_i = \text{Hypertension}_i + \text{Heart\_Disease}_i$$

III. Monetary ($\mathcal{M}$): Metabolic Burden Index.

$$\mathcal{M}_i = \text{BMI}_i + \text{HbA1c\_Level}_i + \text{Blood\_Glucose}_i$$

All features were standardised using Z-score normalisation prior to K-Means.

5.2.2 Optimal Cluster Selection: Elbow Method and Silhouette Analysis

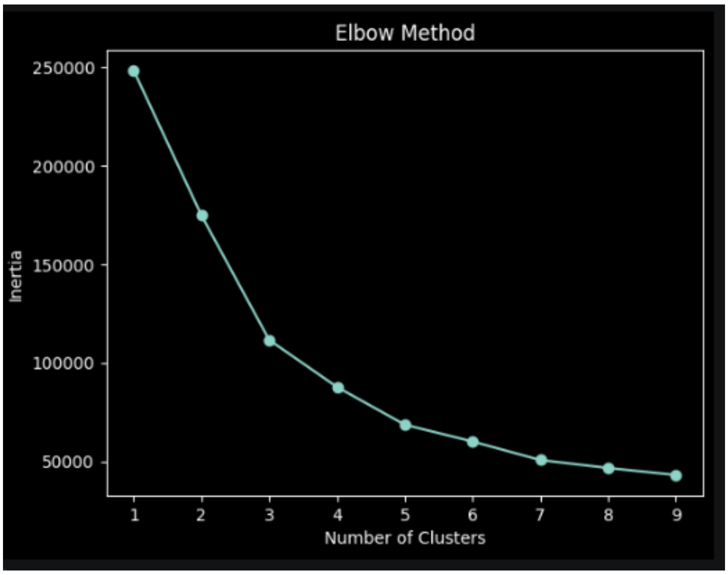


Figure 11: Elbow Method: WCSS (Inertia) vs. number of clusters. A clear inflection appears at $k = 3$–4, beyond which marginal reduction in WCSS becomes negligible.

Silhouette scores were computed for $k = 2, \ldots, 7$. Although the peak silhouette score occurred at $k = 7$, a higher k reduces clinical interpretability. **Therefore, $k = 3$ was selected** to balance statistical performance with interpretability.

5.2.3 K-Means Clustering ($k = 3$)

Cluster	Age	BMI	HbA1c	Glucose	Diabetes Prev.
Cluster 0 (Low Risk)	Low	Moderate	Low	Low	$\approx 5\%$
Cluster 1 (Medium Risk)	Moderate	Moderate	Moderate	Moderate	≈ 10–15%
Cluster 2 (High Risk)	High	High	High	Very High	$> 40\%$

Cluster Profiles:

Cluster 0 — Low Risk. Youngest cohort; low blood glucose and HbA1c below diagnostic thresholds. Likely metabolically healthy.

Cluster 1 — Medium Risk. Middle-aged with moderate glucose and slightly elevated HbA1c. Pre-diabetic or at-risk group requiring monitoring.

Cluster 2 — High Risk. Older individuals with very high blood glucose and HbA1c above the clinical diagnostic threshold. Requires immediate clinical attention.

5.2.4 Cluster Visualisation

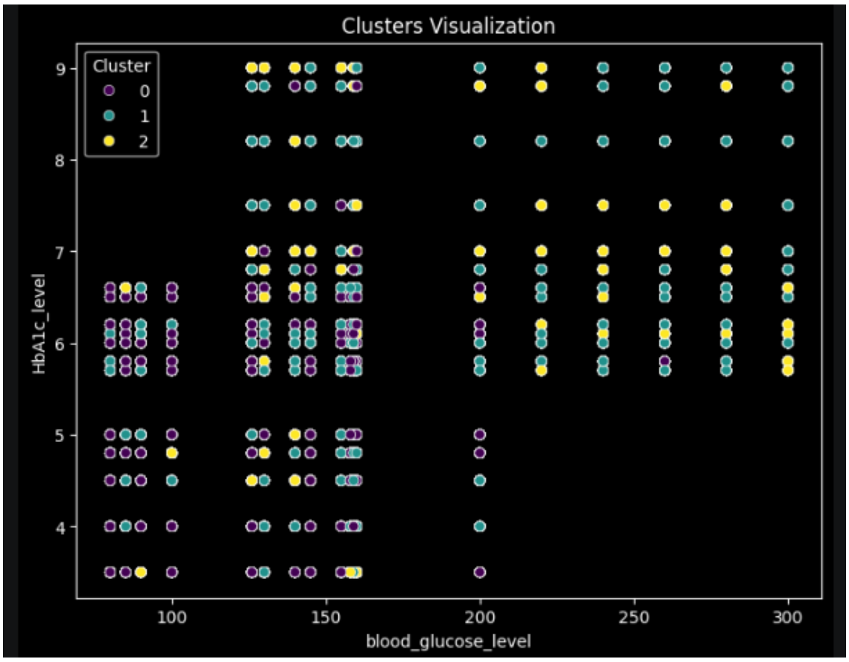


Figure 12: 2D scatter plot of Blood Glucose Level vs. HbA1c coloured by K-Means cluster assignment. Clusters are primarily separated along the HbA1c and glucose axes.

Interpretation: The scatter plot shows how patients are grouped into different clusters based on their blood glucose and HbA1c levels. Each color represents a different cluster identified by the K-Means algorithm. It can be observed that the clusters are mainly separated along the glucose and HbA1c axes, indicating that these two features play a major role in distinguishing patient groups. Patients with higher glucose and HbA1c values tend to fall into the same cluster, representing a higher-risk group, while those with lower values form separate clusters corresponding to lower-risk individuals. Overall, the clustering captures meaningful patterns in the data and helps in identifying groups of patients with similar health conditions.

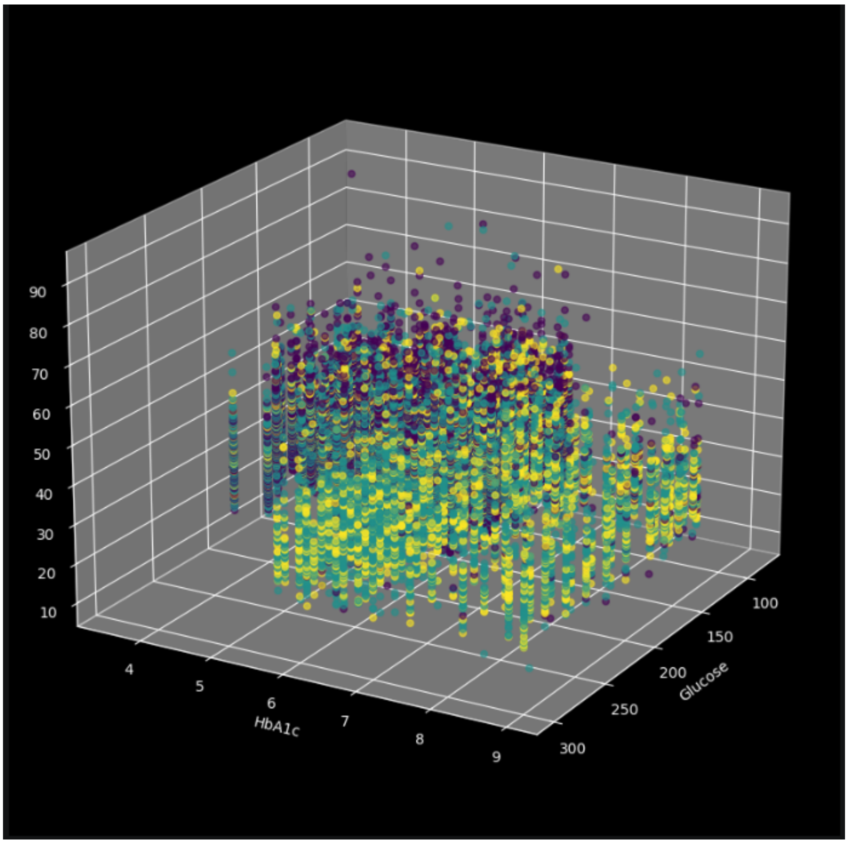


Figure 13: 3D scatter plot (Glucose, HbA1c, BMI) confirming that BMI contributes less to cluster separation than the two glycaemic markers.

Interpretation: The 3D plot shows how patients are grouped based on blood glucose, HbA1c, and BMI. It is clear that most of the separation between clusters happens along the glucose and HbA1c axes, while BMI does not create strong differences between groups. This means that patients with similar BMI values can still belong to different clusters depending on their glucose and HbA1c levels. In simple terms, blood glucose and HbA1c are much more important for identifying different patient groups, while BMI plays a smaller role in distinguishing between them.

5.2.5 PCA-Based Cluster Visualisation

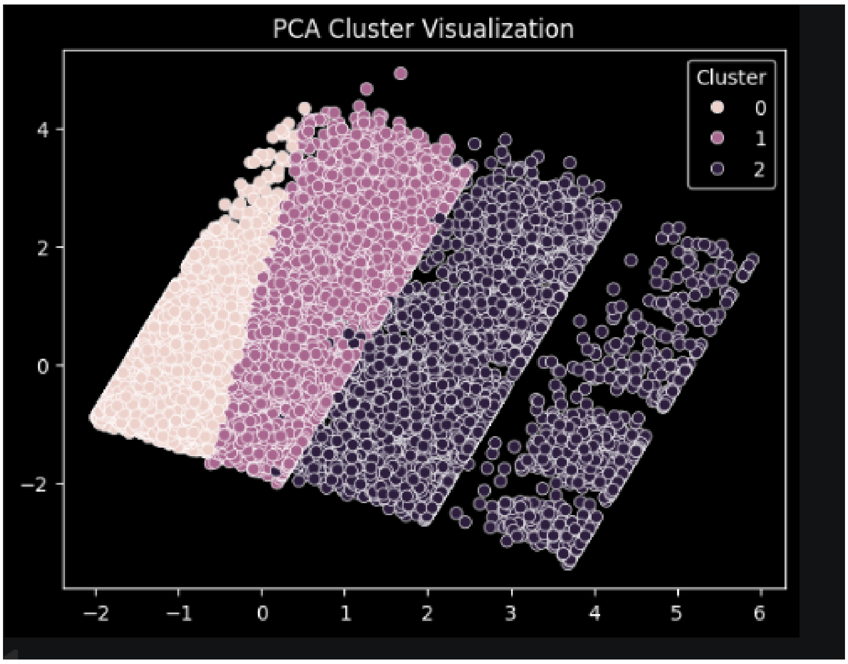


Figure 14: PCA scatter plot (PC1 vs. PC2) with K-Means cluster labels. Clusters form well-separated regions, validating the segmentation quality.

Interpretation: The PCA plot shows how patients are grouped after reducing the data to two main components. Each color represents a different cluster. The clusters are clearly separated from each other, which means the grouping done by the K-Means algorithm is effective. Patients within the same cluster are similar to each other, while those in different clusters are quite different. This clear separation indicates that the model has successfully identified meaningful patterns in the data, helping to divide patients into distinct risk groups.

5.2.6 Hierarchical Clustering

Agglomerative hierarchical clustering with Ward linkage was applied to a random sample of 2,000 patients.

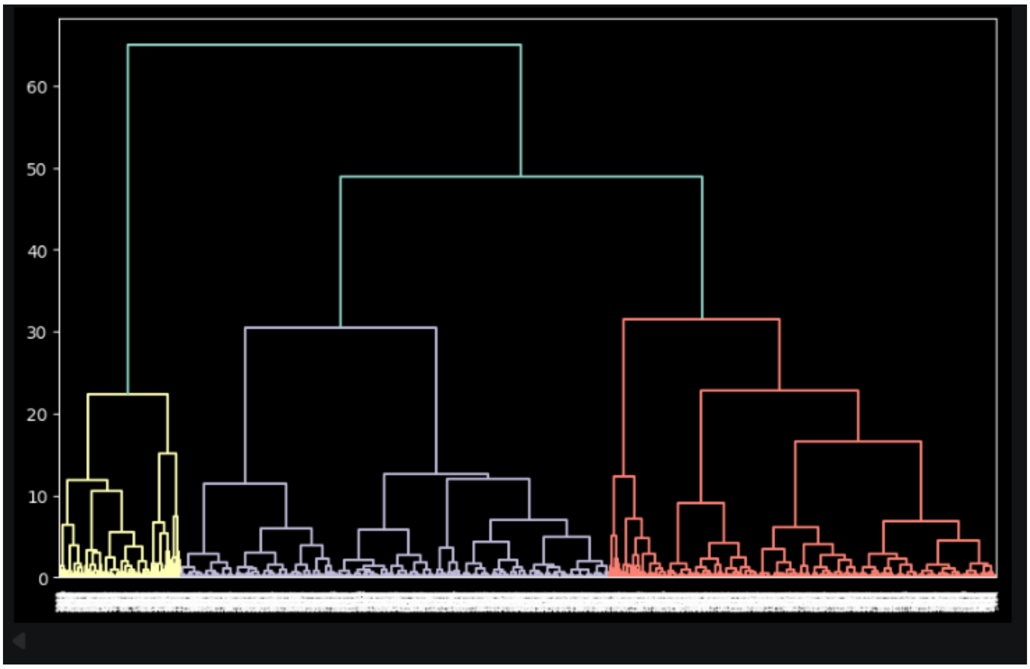


Figure 15: Hierarchical clustering dendrogram (2,000 patient sample, Ward linkage). The structure naturally suggests 2–3 major branches, consistent with the $k = 3$ K-Means result.

Interpretation: The dendrogram shows how patients are grouped step by step based on their similarity. Each branch represents a group of similar patients, and the height at which branches merge indicates how different the groups are. From the plot, we can see that the data naturally forms around 2 to 3 major clusters. This supports the earlier K-Means result where $k = 3$ was chosen. Patients within the same branch are more similar to each other, while those in different branches are more distinct. Overall, the dendrogram confirms that the clustering structure in the data is meaningful and reliable.

5.3 Supervised Classification

5.3.1 Model Overview and Theory

Linear Discriminant Analysis (LDA). LDA finds a linear combination of features that maximally separates classes by optimising the ratio of between-class to within-class variance:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

Logistic Regression. Models the probability of class membership using a logistic function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}$$

Decision Tree and CART. Tree-based models that partition the feature space using recursive binary splits. CART uses the Gini impurity:

$$\mathrm{Gini}(D) = 1 - \sum_{i=1}^{C} p_i^2$$

Support Vector Machine (SVM). Finds the optimal separating hyperplane. With RBF kernel:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)$$

Random Forest. An ensemble of decision trees using bootstrap aggregation:

$$\hat{y} = \arg\max_c \sum_{t=1}^{T} \mathbf{1}[h_t(x) = c]$$

Multi-Layer Perceptron (MLP). A neural network that captures non-linear relationships:

$$a_j^{(l)} = \sigma\left(\sum_i w_{ij}^{(l)} a_i^{(l-1)} + b_j^{(l)}\right)$$

5.3.2 Handling Class Imbalance (SMOTE)

The dataset exhibits significant class imbalance ($\approx 91 : 9$). To address this, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training set. It generates synthetic samples for the minority class by interpolating between nearest neighbours:

$$x_{\text{new}} = x_i + \lambda(x_{nn} - x_i), \quad \lambda \in [0, 1]$$

This improves the model's ability to learn minority class patterns and reduces bias toward the majority class.

5.3.3 Evaluation Metric Definitions

Since the dataset is imbalanced, multiple evaluation metrics are used.

Accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy alone can be misleading in imbalanced datasets.

Precision.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Indicates reliability of positive predictions.

Recall (Sensitivity).

$$\text{Recall} = \frac{TP}{TP + FN}$$

Critical for detecting diabetic patients.

F1-Score.

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Balances precision and recall.

ROC-AUC.

$$AUC = \int_0^1 TPR\, d(FPR)$$

Measures overall classification performance and is robust to class imbalance.

5.3.4 Experimental Setup

Seven supervised models (LDA, Logistic Regression, SVM, Decision Tree, CART, Random Forest, and MLP) were trained using an 80/20 train-test split. SMOTE was applied to the training data to handle imbalance. Feature scaling was applied for models sensitive to input scale (SVM, Logistic Regression, and MLP).

5.3.5 Models Trained

Listing 1: Seven Classification Models

```
models = {
    "LDA":           LinearDiscriminantAnalysis(),
    "Logistic":      LogisticRegression(max_iter=500),
    "SVM":           SVC(kernel='rbf', probability=True),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "CART":          DecisionTreeClassifier(
                         criterion="gini", max_depth=5,
                         min_samples_split=10, random_state=42),
    "Random Forest": RandomForestClassifier(
                         n_estimators=200, max_depth=8, random_state=42),
    "MLP":           MLPClassifier(
                         hidden_layer_sizes=(64,32),
                         max_iter=300, random_state=42)
}
# Note: SVM, Logistic, MLP use scaled data;
```

```
# 	others use unscaled X_train_sm
```

5.3.6 Model Performance Comparison

Model	Accuracy	Precision$_1$	Recall$_1$	F1$_1$	AUC
LDA	0.887278	0.465587	0.830325	0.596628	0.945704
Logistic	0.883714	0.456758	0.835740	0.590687	0.948259
SVM (RBF)	0.883533	0.457858	0.869434	0.599834	0.951907
Decision Tree	0.939592	0.690668	0.721420	0.705709	0.844836
CART (depth 5)	0.871330	0.430024	0.865223	0.574511	0.953112
Random Forest	**0.911864**	**0.539265**	**0.838748**	**0.656463**	**0.961254**
MLP	0.911925	0.540348	0.821901	0.652029	0.964605

Note: Subscript$_1$ denotes performance metrics for the diabetic (minority) class.

Interpretation: The results show that the Multi-Layer Perceptron (MLP) achieves the highest AUC (0.9646), indicating the best overall performance in distinguishing between diabetic and non-diabetic patients. Random Forest also performs strongly with a slightly lower AUC (0.9613) and provides a good balance between precision and recall. SVM achieves the highest recall (0.8694), making it more effective in identifying diabetic cases, which is important for imbalanced datasets. Decision Tree shows high precision and F1-score but lower AUC, indicating less stable performance. Overall, MLP is the best-performing model, while Random Forest and SVM offer strong alternatives depending on the application requirements.

5.3.7 Model Comparison Bar Plot

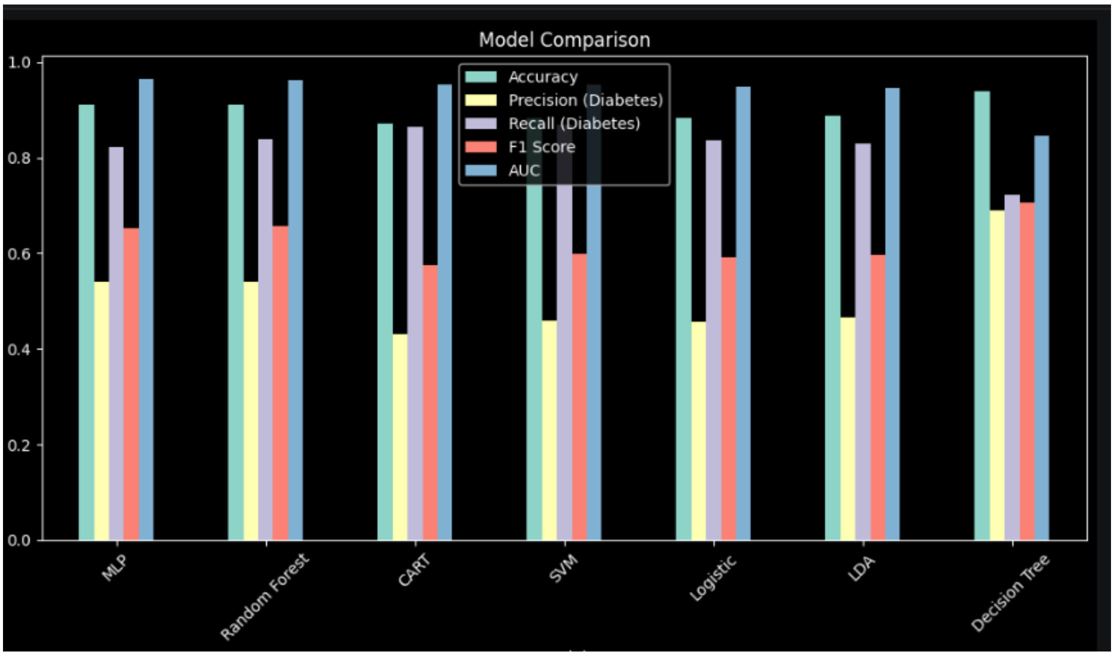


Figure 16: Bar chart comparing Accuracy, Precision (Diabetes), Recall (Diabetes), F1 Score, and AUC across all seven models. Random Forest achieves the best AUC; CART achieves the highest Recall for the minority class.

5.3.8 Combined ROC Curves

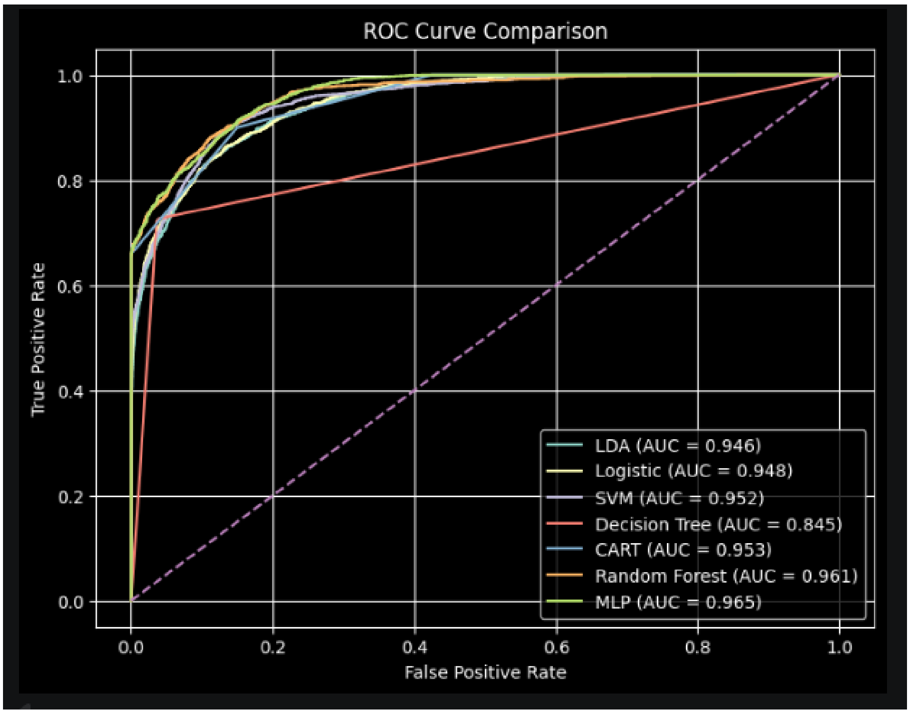


Figure 17: ROC Curve Comparison for all seven models. The dashed line represents the random baseline (AUC = 0.5). MLP achieves the highest AUC (0.965), followed by Random Forest (0.961), CART (0.953), SVM (0.952), Logistic Regression (0.948), LDA (0.946), while Decision Tree shows comparatively lower performance (0.845).

Best Model: Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) achieves the highest AUC (0.965), indicating the best overall discriminative performance among all models. It is closely followed by Random Forest (0.961) and CART (0.953).

The superior performance of MLP can be attributed to its ability to capture complex non-linear relationships in the data. Additionally, the ROC curve shows consistently higher true positive rates across different thresholds.

However, Random Forest remains a strong alternative due to its stability, interpretability, and balanced precision–recall performance under class imbalance.

5.3.9 Random Forest Feature Importance

Feature	Decision Tree	Random Forest
`HbA1c_level`	≈0.521	≈0.419
`blood_glucose_level`	≈0.272	≈0.294
`age`	≈0.082	≈0.097
`gender_encoded`	≈0.057	≈0.084
`age_group_encoded`	≈0.001	≈0.055
`bmi`	≈0.061	≈0.050
`hypertension`	≈0.004	≈0.001
`heart_disease`	≈0.003	≈0.000

HbA1c and blood glucose together account for $> 70\%$ of total feature importance in both tree-based models. Hypertension and heart disease contribute minimally—their signal is already captured through glycaemic features.

6 Key Findings and Clinical Insights

6.1 Summary Statistics

Table 2: Key Clinical Differences Between Diabetic and Non-Diabetic Groups

Feature	Non-Diabetic Mean	Diabetic Mean	Relative Difference
Age (years)	47.43	61.35	+29.3%
BMI (kg/m^2)	28.34	32.11	+13.3%
HbA1c Level (%)	5.39	6.94	+28.7%
Blood Glucose (mg/dL)	132.80	194.01	+46.1%

6.2 Key Insights

1. **Blood glucose and HbA1c** are the strongest predictors, consistent with their clinical diagnostic role (WHO: HbA1c $\geq 6.5\%$, fasting glucose ≥ 126 mg/dL).
2. **Age** acts as a major risk amplifier: older individuals show significantly higher diabetes prevalence compared to younger age groups.
3. **Comorbidities** (hypertension and heart disease) frequently co-occur with diabetes, indicating shared cardiometabolic risk factors.
4. **K-Means clustering** ($k = 3$) successfully identifies low-, medium-, and high-risk patient groups without using class labels. The high-risk cluster shows $> 40\%$

diabetes prevalence.

5. **Association rule mining** reveals meaningful disease patterns, where combinations such as High_HbA1c, obesity, and older age show strong co-occurrence (lift ≈ 1.68).
6. **Multi-Layer Perceptron (MLP)** is the best-performing model with the highest AUC (0.965), indicating superior overall classification performance.
7. **Support Vector Machine (SVM)** achieves the highest recall (≈ 0.87), making it most effective for identifying diabetic patients and reducing false negatives.
8. **Random Forest** provides strong performance (AUC ≈ 0.961, Recall ≈ 0.84) and is useful for feature importance and interpretability.
9. The dataset shows significant **class imbalance** ($\approx 91 : 9$), making metrics such as Recall and AUC more reliable than Accuracy, and requiring SMOTE for balanced learning.

7 Limitations

- **Smoking history was excluded** due to high missingness ($\approx 35.82\%$ "No Info"), which may have omitted a clinically relevant predictor.
- **Cross-sectional design:** Cannot model the longitudinal progression of diabetes risk over time.
- **Generalisability:** The demographic composition is not fully disclosed; findings may not generalise to all populations.
- **Clustering labels** are post-hoc interpretations and should be validated by domain experts before clinical deployment.
- **Self-reported data** may introduce measurement bias in comorbidity variables.

8 Summary

- **Data Preprocessing:** The dataset was cleaned by handling missing values, removing duplicates, and applying feature transformations. Class imbalance ($\approx 91 : 9$) was addressed using SMOTE.
- **Exploratory Data Analysis (EDA):** Clear differences were observed between diabetic and non-diabetic groups. HbA1c and blood glucose showed the most significant variation, making them key indicators.
- **Disease Basket Analysis (Apriori):** Association rules revealed strong co-occurrence patterns. High HbA1c, obesity, and hypertension frequently appeared together with older age, indicating compounding risk factors.

- **Clustering (K-Means):** Patients were segmented into three groups: low-risk, medium-risk, and high-risk. Clusters were mainly separated based on HbA1c and blood glucose levels.
- **Cluster Validation:** PCA and hierarchical clustering confirmed clear and meaningful separation of clusters, validating the choice of $k = 3$.
- **Supervised Models:** Seven models were evaluated. The Multi-Layer Perceptron (MLP) achieved the best performance (AUC = 0.965), followed by Random Forest (0.961).
- **Imbalanced Data Handling:** Recall and AUC were prioritised over accuracy. SVM achieved the highest recall, making it effective for detecting diabetic cases.
- **Feature Importance:** Random Forest analysis showed that HbA1c and blood glucose together contribute more than 70% of the total importance, while other features have relatively smaller impact.
- **Overall Insight:** Combining statistical analysis with machine learning provides an effective framework for early diabetes detection, patient grouping, and risk assessment.

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